

Study Guide: Recursion Depth and Stack Bounds

Recursion is a programming technique where a function calls itself directly or indirectly to solve a problem. Every recursive function requires two fundamental components: a base case that halts the execution, and a recursive step that progresses towards the base case. Without a proper base case, a recursive function will loop indefinitely, consuming stack memory.

The call stack is a stack data structure that stores information about the active subroutines of a computer program. Each recursive call creates a new stack frame containing local variables, arguments, and return addresses. The depth of recursion refers to the number of stack frames currently active. If the depth exceeds the stack bounds, a Stack Overflow exception is raised by the operating system.

To prevent stack overflows, developers can trace execution trees, limit the recursion depth using guards, or convert the algorithm into an iterative version using a loop and an explicit stack array. In python, the `sys.getrecursionlimit()` function returns the current stack depth configuration, which defaults to 1000.